LEARNER GUIDE

The second difference

Quadratics. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Last lesson you noticed that a growing pattern's gaps — the first differences — keep getting bigger. This lesson chases the question that left you with: amid all that growing, is anything staying the same? Take the square numbers 1, 4, 9, 16, 25. The first differences are 3, 5, 7, 9 — growing. Now take the differences of those: 5–3, 7–5, 9–7 — every one is 2. That repeating number is the second difference, and for a quadratic it is always constant. It is the true fingerprint of a quadratic: a linear pattern has a constant first difference, and a quadratic pattern has a constant second difference. You will build difference tables to find that constant, use it to continue a sequence (the second difference gives you the next gap, and the next gap gives you the next term), sort sequences into linear, quadratic or neither, and discover that the second difference is exactly twice the number in front of n^2. There is no calculator today. The question you are exploring is: how can one constant number reveal a hidden quadratic?

What you are learning to notice

- That taking the differences a second time can reveal a constant hiding beneath a growing pattern.
- That a constant first difference means linear, and a constant second difference means quadratic.
- That the second difference is a tool: add it to the last gap for the next gap, then add that to the last term for the next term.
- That the second difference of y = a n^2 is 2 × a — twice the number in front of n^2.

Moving through it, one step at a time

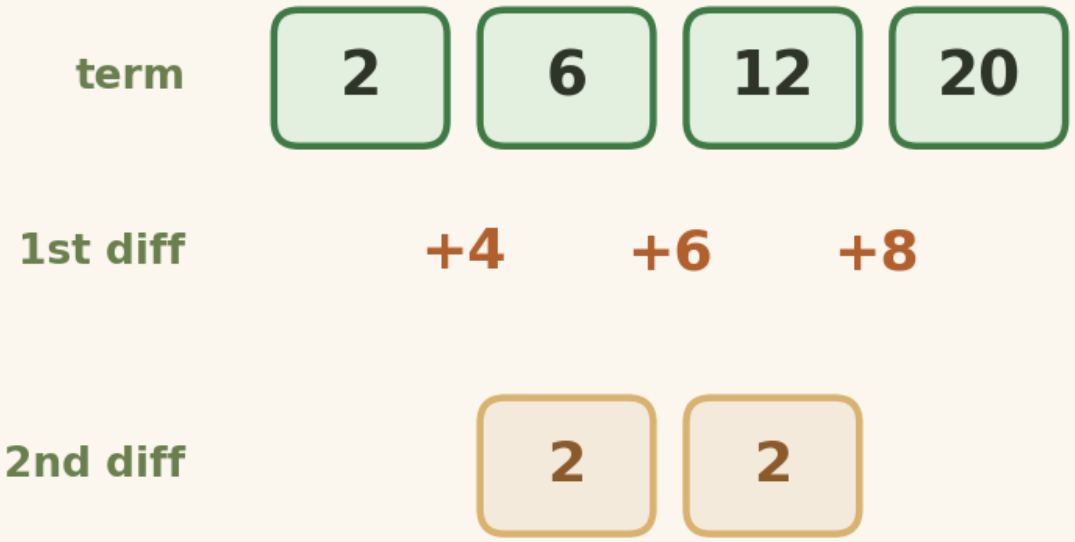
In The fingerprint, reveal a difference table one row at a time and watch the second differences settle to a single number. In Second-Difference Detective, use that constant to continue a quadratic — work back up from the second difference to the next term. In Is it quadratic?, sort sequences into linear, quadratic or neither using the differences. In The second difference and the rule, change the number a in y = a n^2 and see the second difference come out as 2 × a. Build the table each time — terms, then first differences, then second differences.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — is it quadratic?

Build the difference table. If the second differences are all the same, it is a quadratic.

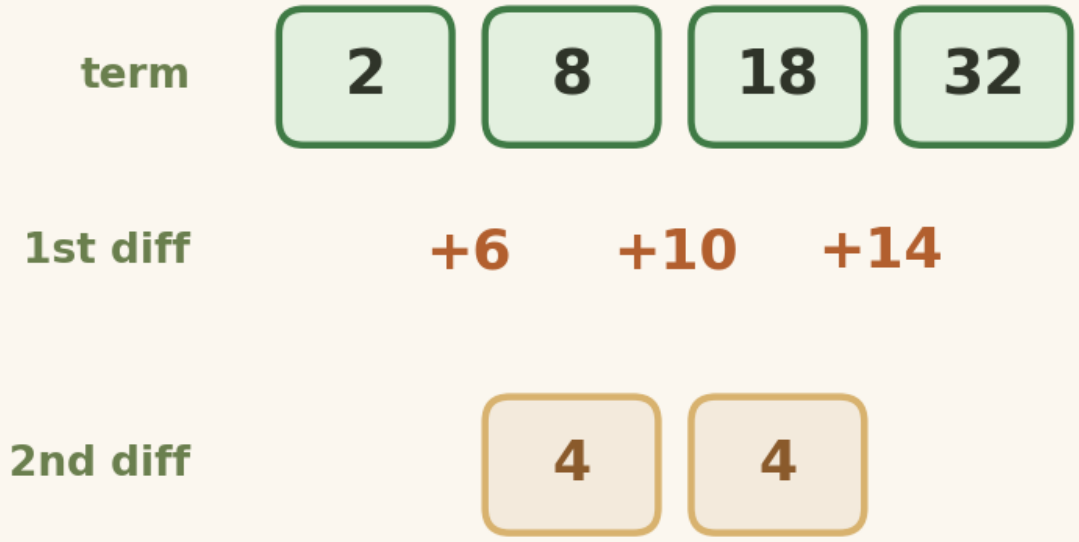


1. the first differences (the gaps)	4, 6, 8
2. the second differences (gaps of the gaps)	2, 2
3. the second difference is constant, so this is	quadratic

quadratic — the second difference is 2

Example 2 — continue using the second difference

The second difference is constant, so use it to reach the next term: it gives the next gap, and the gap gives the next term.



1. the first differences	6, 10, 14
2. the second difference (constant)	4
3. next gap = last gap + 4	14 + 4 = 18
4. next term = last term + next gap	32 + 18

the next term is 50

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (first differences (the gaps between terms, from Lesson 1), squaring a number, and substituting a value into a rule such as $2n^2$), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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